

Hypothesis Testing Exercises

Exercise 1. (Production Line Assembly Time)

The supervisor of a production line believes that the average time to assemble an electronic component is 14 minutes. Assume that assembly time is normally distributed with a standard deviation of 3.4 minutes. The supervisor times the assembly of 14 components, and finds that the average time for completion was 11.6 minutes.

a) What are the appropriate null and alternative hypotheses?

- A. $H_0 : \mu \geq 14$ and $H_A : \mu < 14$
- B. $H_0 : \mu \leq 14$ and $H_A : \mu > 14$
- C. $H_0 : \mu = 14$ and $H_A : \mu \neq 14$
- D. $H_0 : \mu \neq 14$ and $H_A : \mu = 14$

b) Which of the following statements is most accurate?

- A. Unable to reject the null hypothesis at $\alpha \leq 0.10$
- B. Reject the null hypothesis at $\alpha = 0.025$, but not at $\alpha = 0.05$
- C. Reject the null hypothesis at $\alpha = 0.05$, but not at $\alpha = 0.01$
- D. Reject the null hypothesis at $\alpha = 0.008$

Exercise 2. (New Product Output)

The manufacturer of a new product claims that his product will increase output per machine by at least 29 units per hour. A line manager adopts the product on 15 of his machines, and finds that the average increase was only 26 with a standard deviation of 6.2.

a) What are the appropriate null and alternative hypotheses?

- A. $H_0 : \mu = 29$ and $H_A \neq 29$
- B. $H_0 : \mu \geq 29$ and $H_A < 29$
- C. $H_0 : \mu = 29$ and $H_A > 29$
- D. $H_0 : \mu > 29$ and $H_A \leq 29$

b) Using a 5% significance level, which of the following statements is true?

- A. Reject H_0 if t statistic < 1.771
- B. Reject H_0 if t statistic > 2.16
- C. Reject H_0 if t statistic < -1.771
- D. Reject H_0 if t statistic < -2.16

c) The value of the test statistic is:

- A. 1.240
- B. 1.874
- C. -1.240
- D. -1.874

d) Which of the following statements is most accurate?

- A. Reject the null hypothesis at $\alpha = 0.10$
- B. Unable to reject the null hypothesis at $\alpha \leq 0.10$
- C. Reject the null hypothesis at $\alpha = 0.05$, but not 0.10
- D. Reject the null hypothesis at $\alpha = 0.01$, but not 0.05

Exercise 3. (Dentist Survey)

The manufacturer of a certain chewing gum claims that at least 80% of dentists surveyed prefer their type of gum. You decide to test their claim. You find that in a sample of 200 dentists, 74.1% do actually prefer their gum.

a) What are the null and alternative hypotheses for your test?

- A. $H_0 : P \geq 0.80$ and $H_A : P < 0.80$
- B. $H_0 : P = 0.80$ and $H_A : P \neq 0.80$
- C. $H_0 : P \leq 0.80$ and $H_A : P > 0.80$
- D. $H_0 : P > 0.80$ and $H_A : P \leq 0.80$

b) The value of the test statistic is:

- A. 2.086
- B. 1.444
- C. -2.086
- D. -1.444

c) Which of the following statements is most accurate?

- A. Unable to reject the null hypothesis at $\alpha \leq 0.10$
- B. Reject the null hypothesis at $\alpha = 0.05$
- C. Reject the null hypothesis at $\alpha = 0.10$, but not 0.05
- D. Reject the null hypothesis at $\alpha = 0.01$

Exercise 4. (Postal Service Study)

The U.S. Postal Service advertises that at least 63.4% of all third class mail is read by the recipients. An environmental group, concerned about waste generation, wants to test this assertion. They take a sample of 220 households and find that only 58.7% read their third class mail.

- a) What are the null and alternative hypotheses for the environmental group's test?
 - A. $H_0 : P = 0.634$ and $H_A : P > 0.634$
 - B. $H_0 : P = 0.634$ and $H_A : P \neq 0.634$
 - C. $H_0 : P \geq 0.634$ and $H_A : P < 0.634$
 - D. $H_0 : P > 0.634$ and $H_A : P \leq 0.634$
- b) Which of the following statements is most accurate?
 - A. Reject the null hypothesis at $\alpha = 0.10$
 - B. Unable to reject the null hypothesis at $\alpha \leq 0.10$
 - C. Reject the null hypothesis at $\alpha = 0.05$
 - D. Reject the null hypothesis at $\alpha = 0.01$

Exercise 5. (Holiday Shopping Analysis)

A sample of holiday shoppers is taken randomly from a local mall. Forty-nine shoppers were selected and asked what their average spending on gifts would be during the entire holiday season. The point estimate of the population mean was calculated as \$550 and the population standard deviation was calculated as \$92.

- a) Construct a 95% confidence interval of the population mean spending.
- b) Explain how the central limit theorem is used in constructing this interval.

Exercise 6. (Helmet Testing)

A company that manufactures helmets wants to test their helmets to make sure they pass SNELL certification. One of the tests the helmets are required to undergo is a test that simulates real-world impacts. During the test, crash dummies must, on average, experience a force of nearly 300 g in order to pass. The company tests 36 helmets and finds the dummy experienced an average of 298 g with a population standard deviation of 8 g.

- a) Specify the null and alternative hypotheses to determine whether the company's helmets experience, on average, less than 300 g during the SNELL test.
- b) Calculate the value of the test statistic, and find the critical value at a 10% significance level.
- c) At the 10% significance level, can you conclude that the company's helmets experience less than 300 g during the SNELL test? Explain.